

i)

Given

$$r = .001 \text{ m} \quad L = .05 \text{ m} \quad k = 200 \text{ W/m K}$$

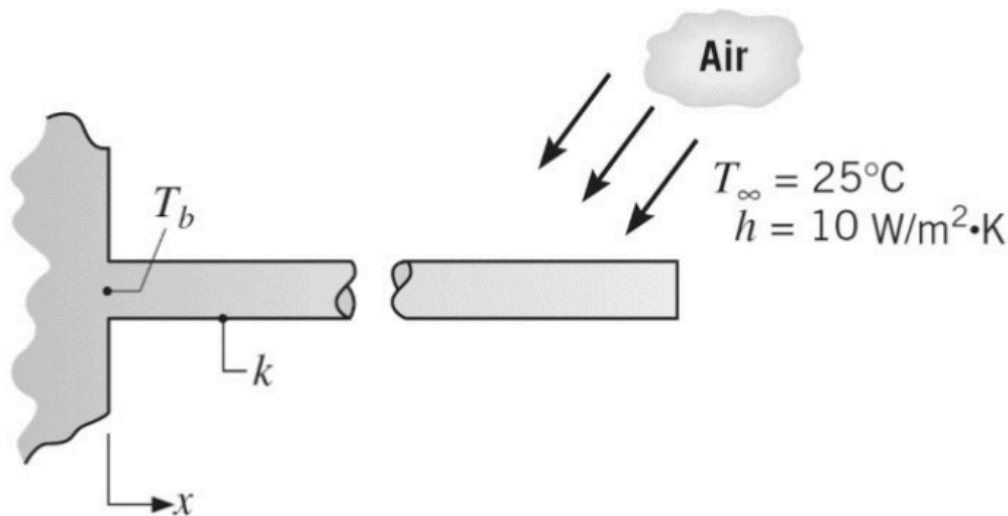
Adiabatic Tip

Schematic below

Find

- $T_{L/2} = ?$, $T_L = ?$ if $T_b = 125^\circ\text{C}$
- ξ_f , η_f by Table 3.4
- η_f by Table 3.5. Compare to part (b)
- $R_f = ?$
- $\eta_o = ?$, R_o
- Resistance network, $T_h = ?$. Compare to no heat sink.

Schematic



Assumptions

SS, 1D heat flow, uniform cross section

Analysis

a) From 3.4:

$$\frac{\theta}{\theta_b} = \frac{\cosh m(L-x)}{\cosh mL}$$

$$m = \sqrt{\frac{(10 \text{ W/m}^2\text{K})(.006 \text{ m})}{(200 \frac{\text{W}}{\text{mK}})(3.14 \text{E-}6 \text{ m}^2)}}$$

$$m = 9.77 \text{ 1/m}$$

$$\theta_{L/2} = (398 \text{ K} - 298) \frac{\cosh(9.77(.05 - .025))}{\cosh(9.77 \cdot .05)}$$

$$\theta_{L/2} = 91.82 \text{ K} = T_{L/2} - T_{\infty} = T_{L/2} - 298 \text{ K}$$

$$T_{L/2} = 389.82 \text{ K} = 116.82^{\circ}\text{C}$$

$$\theta_L = (398 \text{ K} - 298) \frac{\cosh(9.77(0))}{\cosh(9.77 \cdot .05)} = 89.15 \text{ K}$$

$$\theta_L = T_L - T_{\infty} = 89.15 \text{ K}$$

$$T_L = 387.15 \text{ K} = 114.15^{\circ}\text{C}$$

$$\theta = T - T_{\infty}$$

$$m = \sqrt{\frac{hP}{kA_c}}$$

$$P = 2\pi r = 2\pi(.001 \text{ m})$$

$$P = .006 \text{ m}$$

$$A_c = \pi r^2 = \pi(.001 \text{ m})^2$$

$$A_c = 3.14 \text{E-}6 \text{ m}^2$$

$$b) \quad \epsilon_f = \frac{q_f}{q_{nf}}$$

$$q_f = M \tanh(mL)$$

$$M = \sqrt{hPkA_c} \theta_b$$

$$q_{nf} = hA_c \theta_b$$

$$\epsilon_f = \frac{\sqrt{hPkA_c} \theta_b \tanh(mL)}{hA_c \theta_b}$$

$$\epsilon_f = \frac{\sqrt{(0.006 \text{ m})(200 \text{ W/mK})} \tanh(9.77 \cdot .05)}{\sqrt{(10 \text{ W/m}^2\text{K})(3.14 \text{ E-}6 \text{ m}^2)}}$$

$$\epsilon_f = 88.56$$

$$\eta_f = \frac{q_f}{q_{max}}$$

$$q_{max} = hA_f \theta_b$$

$$A_f = PL = (.006 \text{ m})(.05 \text{ m}) = 3 \text{ E-}4 \text{ m}^2$$

$$\eta_f = \frac{\sqrt{hPkA_c} \theta_b \tanh(mL)}{hA_f \theta_b}$$

$$\eta_f = \frac{\sqrt{(0.006 \text{ m})(200 \frac{\text{W}}{\text{mK}})(3.14 \text{ E-}6 \text{ m}^2)} \tanh(9.77 \cdot .05)}{\sqrt{(10 \text{ W/m}^2\text{K})} (3 \text{ E-}4 \text{ m}^2)}$$

$$\eta_f = .927 = 92.7 \%$$

c) From Table 3.5 for pin fin:

$$m = \sqrt{\frac{4h}{KD}} = 10$$

$$\eta_f = \frac{\tanh(mL_c)}{mL_c} = \frac{\tanh(10 \cdot .0505)}{10 \cdot .0505}$$

$$L_c = .05 + \frac{.002}{4}$$

$$\eta_f = .923 = 92.3 \%$$

The expression for pin fin efficiency given in Table 3.5 is a great approximation for η_f .

d)

$$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{(.927)(10 \frac{W}{m^2K})(.006m \cdot .05m)}$$

$$R_f = 359.58 \text{ K/W}$$

e)

$$\eta_o = 1 - \frac{NA_f}{A_{tot}} (1 - \eta_f)$$

$$\eta_o = 1 - \frac{(25)(.006 \cdot .05)}{.0083 \text{ m}^2} (1 - .927)$$

$$\eta_o = .934 = 93.4\%$$

$$\begin{aligned} A_{of} &= S^2 - NA_c \\ &= (.03 \text{ m})^2 - (25)(\pi(.001)^2) \\ &= 8.21 \text{ E-4 m}^2 \end{aligned}$$

$$A_{tot} = NA_f + A_{of}$$

$$A_{tot} = (25)(.006 \cdot .05) + 8.21 \text{ E-4 m}^2$$

$$A_{tot} = .0083 \text{ m}^2$$

$$R_o = \frac{1}{\eta_o h A_{tot}}$$

$$R_o = \frac{1}{(.934)(10 \frac{W}{m^2K})(.0083 \text{ m}^2)} = 12.87 \frac{K}{W} = R_o$$

$$q_o = 25q_f + q_{uf}$$

$$q_f = \frac{\Delta T}{R_f} = \frac{T_b - T_{\infty}}{R_f} = \frac{100 \text{ K}}{359.58 \text{ K/W}}$$

$$q_f = .278 \text{ W}$$

$$q_o = \frac{\Delta T}{R_o} = \frac{T_b - T_{\infty}}{R_o} = \frac{100 \text{ K}}{12.87 \text{ K/W}}$$

$$q_o = 7.77 \text{ W}$$

$$7.77 \text{ W} = 25(.278 \text{ W}) + q_{uf}$$

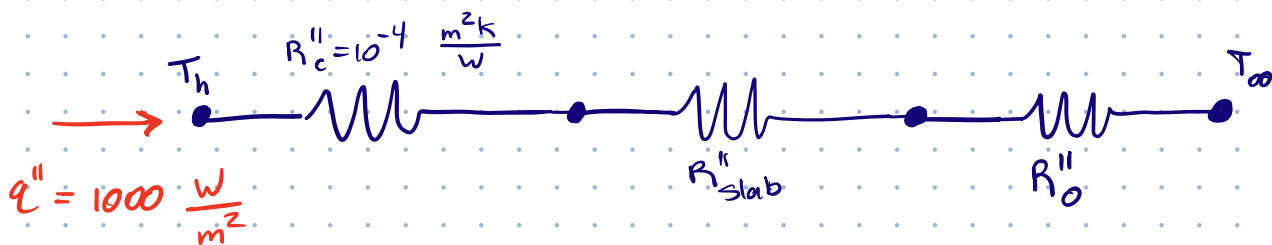
$$\Rightarrow q_{uf} = .817 \text{ W}$$

Heat thru
unfinned surface

$$q_{f, \text{total}} = 25(.278 \text{ W}) = 6.95 \text{ W}$$

Heat thru
fins

f)



$$R_{\text{slab}}'' = \frac{L}{k} = \frac{.02 \text{ m}}{10 \text{ W/mK}} = .002 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_o = 12.87 \frac{\text{K}}{\text{W}}$$

$$R_o'' = R_o s^2 = (12.87 \frac{\text{K}}{\text{W}})(.03 \text{ m})^2 = .012 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$R_{\text{eq}}'' = R_c'' + R_{\text{slab}}'' + R_o'' = .0141 \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$\Delta T = q'' R_{eq}'' = \left(1000 \frac{W}{m^2}\right) \left(0.0141 \frac{m^2 K}{W}\right)$$

$$T_h - T_\infty = 14.1 \text{ K}$$

$$T_h = 312.1 \text{ K} \quad \text{with heat sink}$$

$$T_h = q'' (R_c'' + R_{slab}'' + R_{conv}'') + T_\infty = \left(1000 \frac{W}{m^2}\right) \left(0.0021 \frac{m^2 K}{W} + \frac{1}{10 \frac{W}{m^2 K}}\right) + 298 \text{ K}$$

$$T_h = 400.1 \text{ K} \quad \text{without heat sink}$$

Adding the heat sink reduces the heater temperature by 88°C .